

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

*I **HEMASRINITHA P (192572179)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Hemasrinitha P

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



CSA1708 – ARTIFICIAL INTELLIGENCE TECHNICAL PRESENTATION

REACTIVE AGENT-BASED TRAFFIC CONTROL SYSTEM

NAME: HEMASRINITHA P

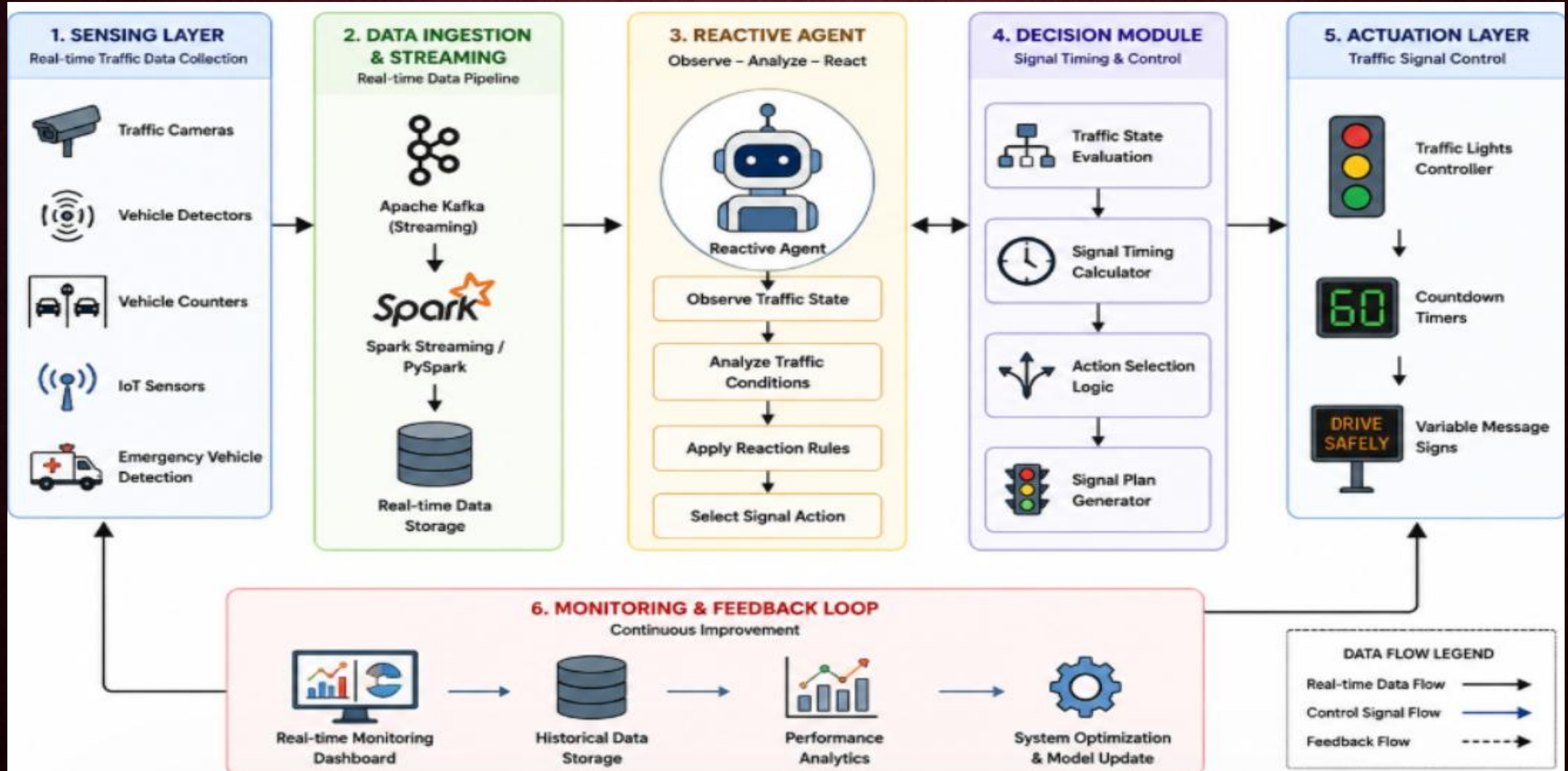
REG.NO.: 192572179



INTRODUCTION

- A Reactive Agent is an intelligent system that responds immediately to changes in its environment.
- In smart traffic control, the agent continuously monitors real-time traffic conditions.
- Traffic inputs include vehicle density, waiting time, lane occupancy, and emergency vehicle presence.
- The agent analyzes current conditions and selects appropriate traffic signal actions.
- The main objective is to reduce congestion, waiting time, and unnecessary traffic delays.

SYSTEM ARCHITECTURE



REACTIVE AGENT WORKING

- The agent continuously receives the latest traffic information from the environment.
- It compares current traffic conditions with predefined response rules.
- When traffic density increases, the agent can extend the green signal duration.
- When traffic decreases, the agent can reduce unnecessary green time and adjust signal priority.
- The agent repeatedly observes and reacts without requiring a long-term prediction model.



APPLICATIONS



- **Smart Traffic Signals** – Dynamically adjusts signal timings based on real-time vehicle density.
- **Emergency Vehicle Priority** – Detects ambulances and fire vehicles and provides priority through signal control.
- **Highway Traffic Management** – Monitors traffic flow and reacts to congestion on major roads.
- **Intelligent Intersection Control** – Coordinates multiple traffic signals to improve vehicle movement at intersections.
- **Smart City Transportation** – Integrates real-time traffic data to support efficient and automated urban mobility.



CONCLUSION

A Reactive Agent-Based Traffic Control System continuously monitors real-time traffic conditions and responds immediately to changing vehicle flow. By processing streaming data from sensors and cameras, the agent dynamically adjusts traffic signals to reduce congestion and waiting time. It can also provide priority to emergency vehicles and improve traffic movement at intersections. The integration of AI agents, IoT sensors, and real-time data processing enables efficient and intelligent traffic management.

THANK YOU